

Calculate the value of the phase difference between the voltage U

at the generator terminals:

$$S_n = 62.5 \text{ MVA}$$

$$U_n = 10.5 \text{ kV}$$

$$\cos \phi = 0.8$$

$$x_d = x_q = 194\%$$

$$x_d' = x_q' = 15\%$$

Transformer:

$$S_{nt} = 65 \text{ MVA}$$

$$P = 121 / 10.5$$

$$U_k = 10\%$$

$$\text{System: } U_s = 118 \text{ kV}$$

$$I_o = 409 \text{ A}$$

①

The synchronizing provides switch on HV side of transformer:

$$\text{Transformer reactance: } X_t = U_k \cdot \frac{U_{nt}^2}{S_{nt}} \cdot \frac{S_n}{U_n^2} = 0.1 \cdot \frac{(10.5 \cdot 10^3)^2}{65 \cdot 10^6} \cdot \frac{62.5 \cdot 10^6}{(10.5 \cdot 10^3)^2} \rightarrow$$

$$[X_t = 0.096]$$

$$\text{total reactance for sync. } X = x_d' + X_t = 0.15 + 0.096 = 0.246$$

$$[X = 0.246] \text{ total}$$

$$\text{Relative voltage: } u = \frac{U_s}{U_n} \cdot \frac{1}{P} = \frac{118}{10.5} \cdot \frac{10.5}{121} = 0.975$$

$$[u = 0.975]$$

$$\text{Generator base current: } I_{base} = \frac{S_n}{\sqrt{3} \cdot U_n} = \frac{62.5 \cdot 10^6}{\sqrt{3} \cdot 10.5 \cdot 10^3} = 3436.6 \text{ A}$$

In our problem our value for I_o is the current on the HV side for converting it to the generator side: $I_{gen} = I_{HV} \cdot p \rightarrow$

$$I_{gen} = I_{HV} \cdot p = 409 \cdot \frac{121}{10,5} = 4713.28 \text{ A}$$

in power units related to the base we get

$$i_0 = \frac{I_{gen}}{I_{base}} = \frac{4713.28}{3436.6} = 1.372 \text{ p.u.}$$

For finding phase difference:

$$i_0 = \frac{U}{X_d' + X_t} \cdot 2 \cdot \sin \frac{\delta}{2} \quad \text{then,}$$

$$\sin \frac{\delta}{2} = \frac{i_0 \cdot X_{total}}{2 \cdot U} = \frac{1,372 \cdot 0,246}{2 \cdot 0,975}$$

$$\sin \frac{\delta}{2} = 0,1731$$

$$\left[\delta = 19.932^\circ \right]$$

② The output of alternator with a damper:

$$I_{HV} = ?$$

a) The alternator was in no-load before short circuit.

in no-load we take $e_{q0}'' = e_0 = u_0 = 1$

We can use values from before problem, that we have already

$$\text{then, } i_{k0}'' = \frac{e_{q0}''}{x_d'' + x_t}$$

$$e_{q0}'' = 1$$

$$x_d'' = 0,15$$

$$x_t = 0,096$$

$$i_{k0}'' = \frac{1}{0,246} = 4,06 \text{ p.u.}$$

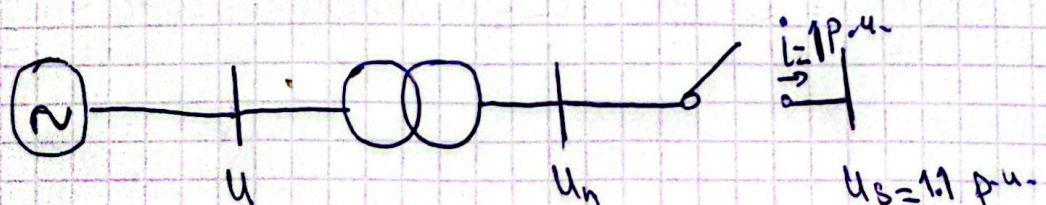
With real values current will be on the HV side:

$$I_{HV} = i_{k0}'' \cdot \frac{S_n}{\sqrt{3} U_n} \cdot \frac{1}{P} = 4,06 \cdot \frac{62,5 \cdot 10^6}{\sqrt{3} \cdot 10^5} \cdot \frac{10,5 \cdot 10^3}{121 \cdot 10^3} \Rightarrow$$

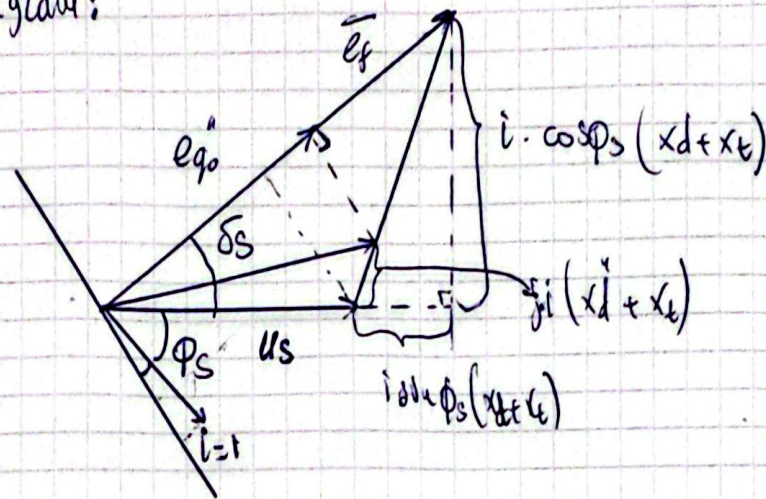
$$[I_{HV} = 1210,76 \text{ A}]$$

b) Alternator supplied nominal load at the power system
Voltage $u_s = 1,1 \text{ p.u.}$ before short-circuit

For this problem: $\begin{cases} i = 1 \text{ p.u.} \\ u_s = 1,1 \text{ p.u.} \end{cases}$



Phasor diagram:



$$i = 1 \text{ p.u.}$$

$$\cos \phi_s = 0,8 \quad \sin \phi_s = 0,6 \quad \phi_s = 36,87^\circ$$

Then phase difference from the phasor diagram will be:

$$\delta_s = \arctg \frac{i \cdot \cos \phi_s (x_d + x_t)}{U_s + i \sin \phi_s (x_d + x_t)} = \frac{1 \cdot 0,8 \cdot 2,036}{1,1 + 1 \cdot 0,6 \cdot 2,036} \Rightarrow$$

$$\left[\delta_s = \arctg \frac{1,6288}{2,3216} = 35,05^\circ \right]$$

Then load current on the d' -axis will be

$$i_{d0} = i \cdot \sin (\delta + \phi) = 1 \cdot \sin (36,87 + 35,05) = 0,95$$

$$\text{Then, } \left[e_{q0}^* = U \cdot \cos \delta + i_{d0} \cdot x_d' \Rightarrow 1,1 \cdot \cos (35,05) + 0,95 \cdot 0,15 = 1,043 \text{ p.u.} \right]$$

$$\text{Short circuit current will be: } i_{k0}^* = \frac{e_{q0}^*}{x_d' + x_t} = \frac{1,043}{0,246} = 4,24 \text{ p.u.}$$

$$\text{Real Value will be: } I_{HV} = i_{k0}^* \cdot \frac{S_n}{\sqrt{3} U_n} \cdot \frac{1}{P} = \left[I_{HV} = 1264,45 \text{ A} \right]$$

$$= 4,24 \cdot 298,22 = 1264,45 \text{ A}$$